

# Zelalem Fissiha Demssie

Fullstack Developer • Data Analyst • ML Researcher • Lecturer

Ethiopian Public Service University — College of Computing and Digitalization
Department of Information Systems — Addis Ababa, Ethiopia

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Portfolio | GitHub | LinkedIn | ORCID

## Professional Summary

Results-oriented full-stack developer, data analyst, machine learning researcher, and university lecturer with over six years of experience in software engineering, data analysis at higher education, and public-sector digital transformation. Builds scalable web platforms with React, Next.js, Node.js, and PostgreSQL, and applies Python (TensorFlow, Scikit-learn, Pandas, NumPy) to AI, ML and data analysis problems. Combines strong skills in requirements analysis, system design, curriculum development, and applied research with a record of leading digital projects that deliver practical academic and public-service impact.

## Key Achievements

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|---|---|
| **Full Stack** | Developed dynamic fullstack website with integrated database and backend functionality |
| **Mentoring** | Supervised 40+ capstone software projects and delivered community IT training. |
| **Research** | Published ML research on LEACH optimization in *Comput. Intell. Neurosci.* (2022). |
| **Academics** | MSc Software Engineering (GPA 3.79/4.00); BSc Computer Science (GPA 3.76/4.00). |
| **Leadership** | Led national outcome-based curricula in Information Systems and Software Engineering. |
| **Recognition** | 1st Place, Best Bank-Queuing Algorithm Design Award, Dire Dawa University (2017). |

## Professional Experience

2025 – Present

**Fullstack Developer — Ethiopian Public Service University, Addis Ababa, Ethiopia**

Design and develop scalable web applications and institutional platforms using React, Next.js, Node.js, PostgreSQL, and RESTful APIs. Architect database schemas and translate user requirements into dynamic, scalable, reliable, maintainable systems that support university operations.

**Data Analyst —** Analyze and visualize institutional, business, software, and educational datasets using Python (Pandas, NumPy, Matplotlib), Excel, and Power BI to identify trends, detect anomalies, monitor performance, and support evidence-based decision-making and data-driven strategic planning.

**ML Researcher and Lecturer —** Conduct applied research in AI, machine learning and educational data mining, and led the ML-driven Integrated Lecture–Lab Framework to strengthen practical programming instruction. Teach undergraduate and graduate courses in Information Systems, AI, Database Systems, Software Engineering, and Computer Networks, and develop outcome-based curricula. Supervised 40+ capstone software projects, led quality audits, and delivered community IT training.y

2017 – 2025

**Lecturer and IT Officer — Ambo University, Ambo, Ethiopia**

Taught undergraduate courses in computer science, Python Programming, AI, database systems, and computer networks. Managed and delivered institutional software projects across the full lifecycle (requirements gathering, analysis, design, and development) and provided technical QA, deployment, and system documentation.

2016 – 2017 **Digital Marketing Officer —** Welkesa and Chilalo Media, Addis Ababa, Ethiopia
Managed websites and blogs, published digital content, and executed online communication initiatives.

## Education

2025 – 2026 **MSc in Project Management** (ongoing), Florida Southeast University (FUSE), USA

2024 – 2025 **Higher Diploma Program in Teaching in Higher Education (HDP)**, Ambo University

2019 – 2021 **MSc in Software Engineering**, Bahir Dar University (BiT), Ethiopia. GPA: 3.79/4.00.
*Thesis: Class-Imbalance Software Defect Prediction Model Using Machine Learning.*

2012 – 2016 **BSc in Computer Science**, Ambo University, Ethiopia. GPA: 3.76/4.00 (Project: A+).
*Project: Web-Based Drug Store Management System for Ethiopian Red Cross Pharmacy.*
*Award: 1st Place, Best Algorithm Design Award (Bank Queuing System Algorithm).*

## Technical Skills

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|---|---|
| **Languages** | Python, TypeScript, JavaScript, Java, Kotlin, Swift, C++, C, R, Rust |
| **Front-End** | React, Next.js, Bootstrap, Tailwind CSS, Angular, HTML5, CSS3 |
| **Back-End** | Node.js, NestJS, Express.js, RESTful APIs, PHP, Django |
| **Software Eng.** | SDLC, Requirements Engineering, Software Testing, QA, Git/GitHub, Agile/Scrum |
| **Data Analysis** | Pandas, NumPy, Matplotlib, Excel, Power BI, Exploratory and Statisical Data Analysis |
| **ML and AI** | Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, OpenCV |
| **Databases** | PostgreSQL, MySQL, MongoDB, SQLite, Firebase, Oracle |

## Selected Projects & Community Service

2025 **Deep Learning Framework for Automated Software Defect Prediction** (Lead Developer). Developed a deep learning–based framework that predicts software defects in institutional systems, improving quality assurance and reducing manual testing effort.

2025 **Integrated Lecture–Lab Framework using ML** (Lead Developer). Developed a machine learning–driven framework that enhances practical programming instruction.

2024 **Digital Literacy & ICT Integration Training** (Project Coordinator). Coordinated a capacity-building program that equipped high school teachers to use technology in teaching and learning.

2023 **Student Academic Performance Prediction System** (Lead Developer). Built a predictive machine learning pipeline that identifies at-risk students early, enabling timely intervention.

## Research & Publications

2022 Demssie, Z. F., *et al.* (2022). "Optimization of LEACH Protocol in Wireless Sensor Network Using Machine Learning." *Computational Intelligence and Neuroscience*, Article 5393251. doi:10.1155/2022/5393251

2020 Demssie, Z. F. (2020). "Software Defect Prediction Model Using Filter Feature Selection and Support Vector Machine Algorithms on Highly Class-Imbalanced Software Dataset." Manuscript.

## Leadership & Contributions

2025 – 2026 **Curriculum Development Team Leader** - Led multidisciplinary teams at EPSU in designing and reviewing outcome-based curricula for Computer Science, Software Engineering, and Intelligent Systems, aligning learning outcomes with national competency standards and contributing to Ethiopia's national curriculum frameworks.

2022 – 2025 **Quality Assurance & Auditing Manager**, Ambo University. Managed audits of Computer Science course delivery and ensured pedagogical compliance of examinations.

2024 – 2025 **National Exit Exam Task Force Leader**, Ambo University. Coordinated the planning and implementation of the National Exit Examination.

2023 – 2025 **Industrial Project Advisor**, Ambo University. Supervised 40+ final-year industry-based software capstone projects and delivered community IT training programs.

2023 – 2025 **Course Module Developer**, Ambo University. Authored course modules for Design & Analysis of Algorithms, Fundamentals of Software Engineering, and Object-Oriented Software Engineering (OOSE).

## Development & Certifications

2025 **Certificate of Big-data** Successfully completed the online training in Big-data, GIS, Data Visualization, with a focus on Power BI for humanitarian data analysis..

2025 **Udacity Nanodegrees:** AI Fundamentals, Programming Fundamentals, Data Analysis Fundamentals, and Android Developer Fundamentals.

2025 **Peace Corps USA :** 7-Month Virtual Classroom English Methodology Program.

2025 **Ministry of Education (e-SHE):** OEX101: Building in Open edX Studio.

2024 **ASU Online / e-SHE:** Master Class for Teaching Online.

2017 **Algorithm Design Award:** Awarded 1st place in the 2nd Annual Programming Week for designing an optimal Bank Queuing System algorithm held at Dire Dawa University.

## Research Interests

**Core focus:** Artificial Intelligence, Machine Learning, and Deep Learning; Educational Data Mining; AI and Data Governance; Business Data Analytics and Business Intelligence; Software Quality Assurance; Wireless Sensor Networks; Software Development and Business Analysis; and ICT for Public Sector Digital Transformation.

## Professional Skills & Languages

**Competencies:** Clear written and verbal communication, strategic leadership, analytical problem-solving, team collaboration, Agile project management, and rapid adaptability.

**Languages:** Amharic (native) • English (fluent) • Afaan Oromo (good)

## References

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